MIL-E-1/1602B(NAVY)
4 September 1970
SUPERSEDING
MIL-E-1/1602A(NAVY)
11 September 1968

MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, NUMERICAL INDICATOR

## TYPE B5961

**The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.**

DESCRIPTION: Cold cathode, gas filled, glow discharge (neon).

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Ebb | Rp | Ik | TA | Alt. |
|---|---|---|---|---|---|
| Unit: | Vdc | ohm | mA | °C | |
| Maximum: | --- | --- | 4.5 | +70 | (See note 1) |
| Minimum: | 170 | --- | --- | -65 | --- |
| TEST CONDITIONS: | 170 | 12K +1% | --- | --- | --- |

GENERAL:

Qualification - Not Required





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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSP. LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MIN. | MAX. | |
| | Quality conformance inspection, part 1 | | | | | | | |
| 3347 | Ionization voltage (1) (with illumination) | Ambient illumination 5-50 footcandles | | ---- | Ez | ---- | 170 | Vdc |
| 3333 | Uniformity of cathode glow | Ambient illumination 5-50 footcandles | | ---- | ---- | ---- | ---- | ---- |
| 1256 | Electrode current (cathode) | | ---- | ---- | Ik | 2.5 | 4.0 | mA |
| | Quality conformance inspection, part 2 | | | | | | | |
| ---- | Light output | Light individual cathodes (Note 2) | ---- | ---- | ---- | 0.35 | ---- | ftC |
| ---- | Alinement digits | (Note 3) | ---- | ---- | ---- | ---- | +3 | degrees |
| 1041 | Shock test | G=250; duration 1.0 +.5 ms. No voltage (Note 4) | ---- | ---- | ---- | ---- | ---- | ---- |
| 1031 | Variable-frequency vibration | (Notes 4 and 5) | ---- | ---- | ---- | ---- | ---- | ---- |
| | Quality conformance inspection, part 3 | | | | | | | |
| ---- | Life test (1) | Sequential counting; 1 Hz; group C (Note 6) t=1000 hours minimum | ---- | ---- | ---- | ---- | ---- | ---- |
| ---- | Life-test (1) end points | Character recognition (Note 8) | ---- | ---- | ---- | ---- | ---- | ---- |
| | | Cathode current | ---- | ---- | Ik | 2.0 | 4.5 | mA |
| | | Light output | ---- | ---- | ---- | 0.3 | ---- | ftC |
| ---- | Life test (2) | One numeral cathode glowing steadily group C (Note 7); t=1000 hours minimum | ---- | ---- | ---- | ---- | ---- | ---- |



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSP. LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MIN. | MAX. | |
| | Quality conformance inspection, part 3 (Continued) | | | | | | | |
| ---- | Life-test (2) end points | Character recognition (Note 8) numeral under test | ---- | ---- | ---- | ---- | ---- | ---- |
| | | Cathode current, numeral under test | ---- | ---- | Ik | 2.0 | 4.5 | mA |
| | | Light output, numeral under test | ---- | ---- | ---- | 0.3 | ---- | ftC |

NOTES:

1. See reduced pressure (altitude) rating, and altitude, maximum peak voltage.
2. Light output of each numeral is measured with a foot-candle (ftC) meter and an eye-corrected photronic cell. The entire cell area is illuminated and is coupled to the tube by a light-tight cylinder so that the face of the cell is 1-1/2 inches from the tube base.
3. The centerline of the numerals drawn through the top of the numeral support posts, shall be within the tolerance specified herein, in relation to a line drawn on the base of the tube through pins 6 and 12.
4. No permanent shorts or opens will be permitted.
5. On each plane the ten cathodes shall be sequentially energized at the test conditions.
6. Tubes shall be energized with a duty cycle greater than 90 percent.
7. (a) Life-test group C, sampling plan, establishment of prerelease eligibility and prerelease provisions.

   Life-test group C, sampling plan:

   (1) The first sample size = 3
   (2) The first cumulative sample size = 3
   (3) The second sample size = 3
   (4) The second cumulative sample size = 6

   Life-test group C, establishment of prerelease eligibility: Minimum, 15 tubes

   Prerelease Provisions: Replace all reference to the current 4 tube sample with current 3 tube sample.

   Replace all reference to the last 16 with the last 15.

   (b) One tube shall have the top numeral cathode glowing steadily. One tube shall have the bottom numeral cathode glowing steadily. One tube shall have the "0" numeral cathode glowing steadily.



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8. Glow coverage shall be such as to insure character recognition.

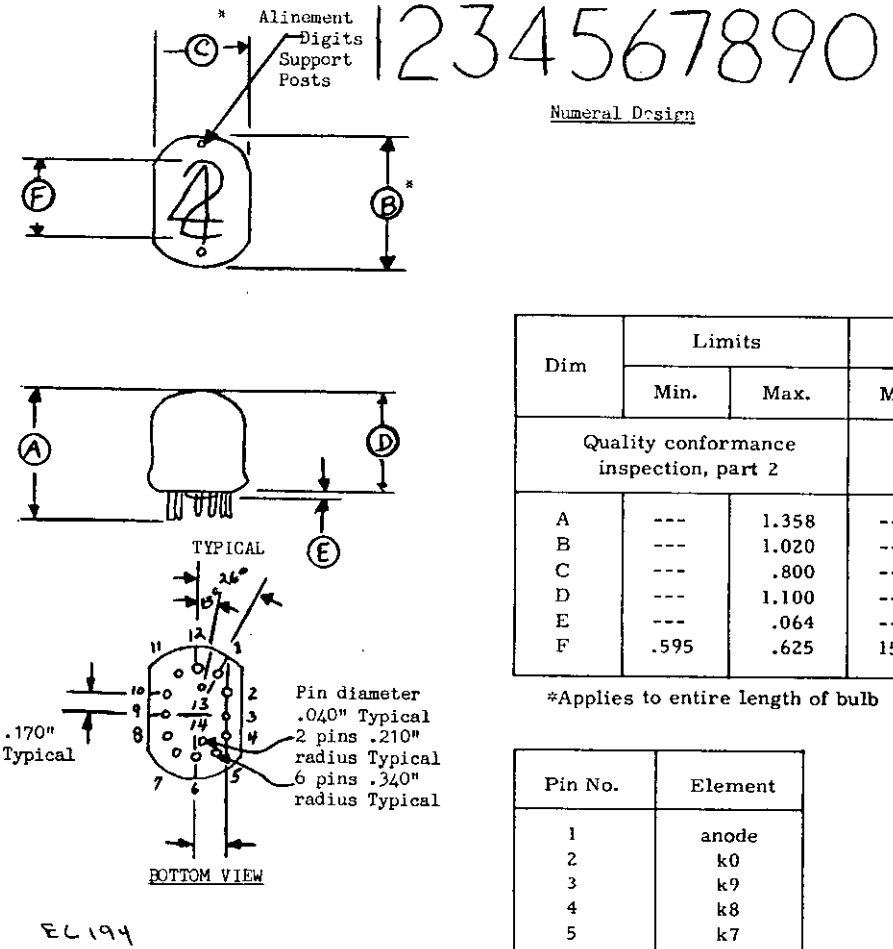
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N824-39)



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| Dim | Limits | | Limits | |
|---|---|---|---|---|
| | Min. | Max. | Min. | Max. |
| Quality conformance inspection, part 2 | | | mm | |
| A | --- | 1.358 | --- | 34.39 |
| B | --- | 1.020 | --- | 25.91 |
| C | --- | .800 | --- | 20.32 |
| D | --- | 1.100 | --- | 27.94 |
| E | --- | .064 | --- | 1.63 |
| F | .595 | .625 | 15.11 | 15.88 |

*Applies to entire length of bulb

| Pin No. | Element |
|---|---|
| 1 | anode |
| 2 | k0 |
| 3 | k9 |
| 4 | k8 |
| 5 | k7 |
| 6 | k6 |
| 7 | k5 |
| 8 | k4 |
| 9 | k3 |
| 10 | k2 |
| 11 | k1 |
| 12 | int con |
| 13 | int con |
| 14 | int con |

FIGURE 1. Outline drawing of tube type B5961



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